



NATIONAL TEACHERS COLLEGE

Software Design & Analysis Document (SDAD)

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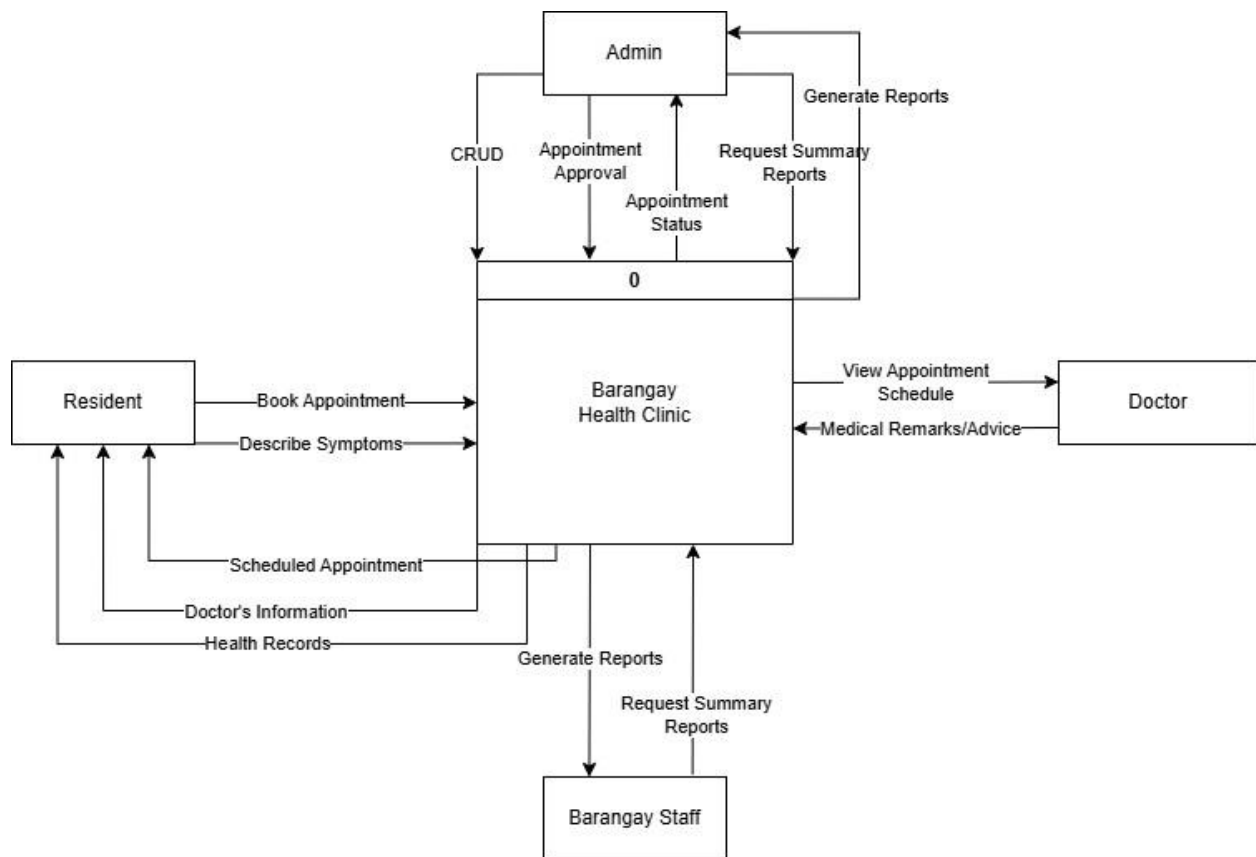
Submitted To:

Justin Louise R. Neype



Process Architecture

LEVEL 0

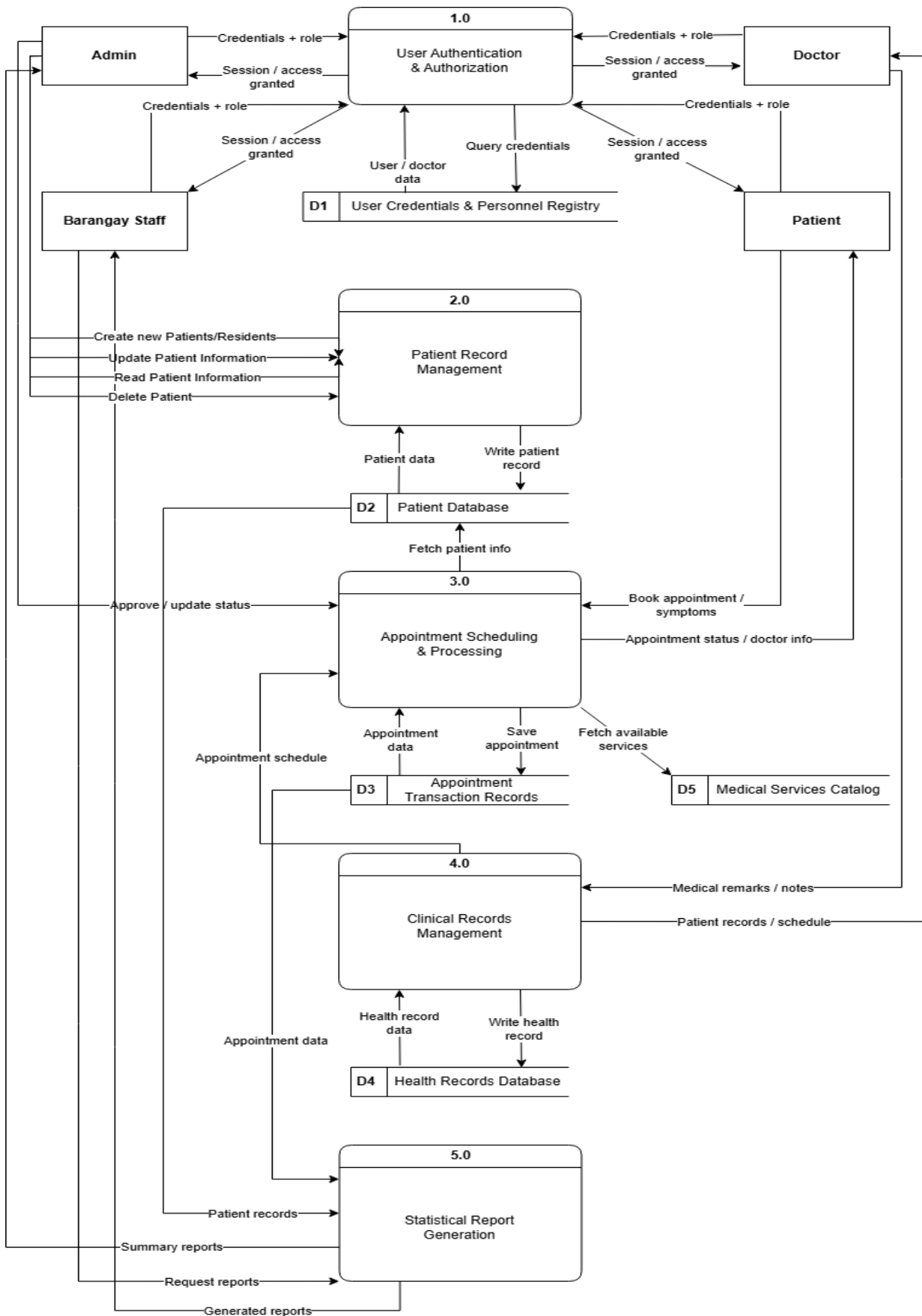


DFD The Level 0 Data Flow Diagram (DFD) of the Barangay Health Clinic System presents the overall flow of information between the system and its external entities, namely the Admin, Resident, Doctor, and Barangay Staff. The diagram shows the Barangay Health Clinic System as a single main process that handles all system operations. Residents can book appointments, describe symptoms, and receive appointment schedules, doctor information, and health records from the system. Doctors can view appointment schedules and provide medical remarks or advice. The Admin is responsible for CRUD operations, appointment approval, monitoring appointment status, and requesting summary reports. Meanwhile, the Barangay Staff can request and generate reports for monitoring purposes. Overall, the Level 0 DFD provides a general overview of how users interact with the Barangay Health Clinic System and how information flows throughout the system.

LEVEL 1



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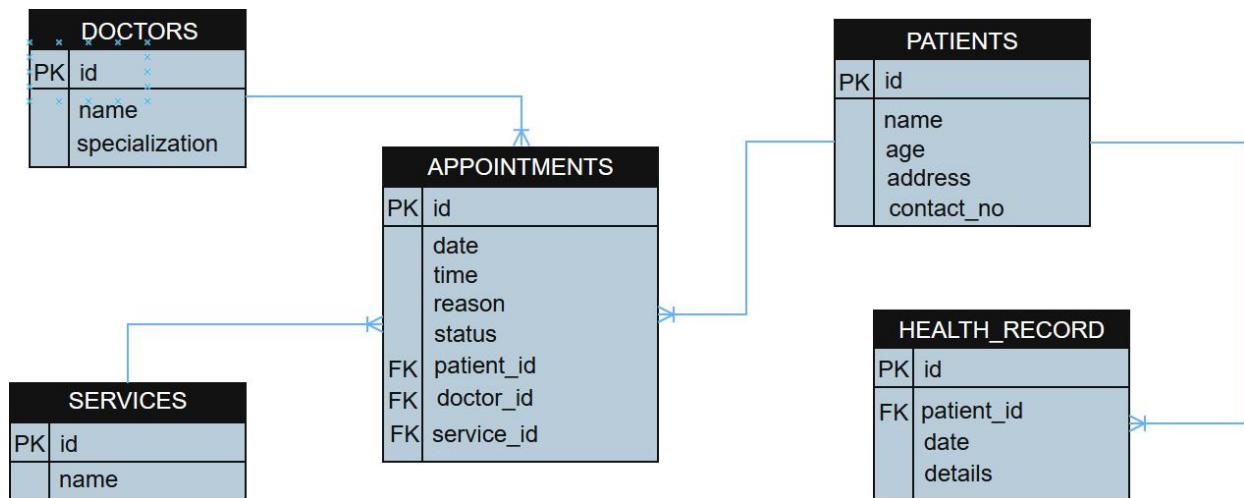
Description:

The Level 1 Data Flow Diagram (DFD) of the Barangay Health Clinic System shows the major processes and how data moves within the system. It breaks down the main system into five core processes: User Authentication and Authorization, Patient Record Management, Appointment Management, Medical Records Management, and Report Generation.

The diagram illustrates how the Admin, Barangay Staff, Doctor, and Patient interact with the system, while data is stored and retrieved from the different databases such as Users, Patients, Appointments, Health Records, and Services. It also shows how the system manages user access, patient records, appointment scheduling, medical information, and report generation.

Data Architecture

ERD DIAGRAM



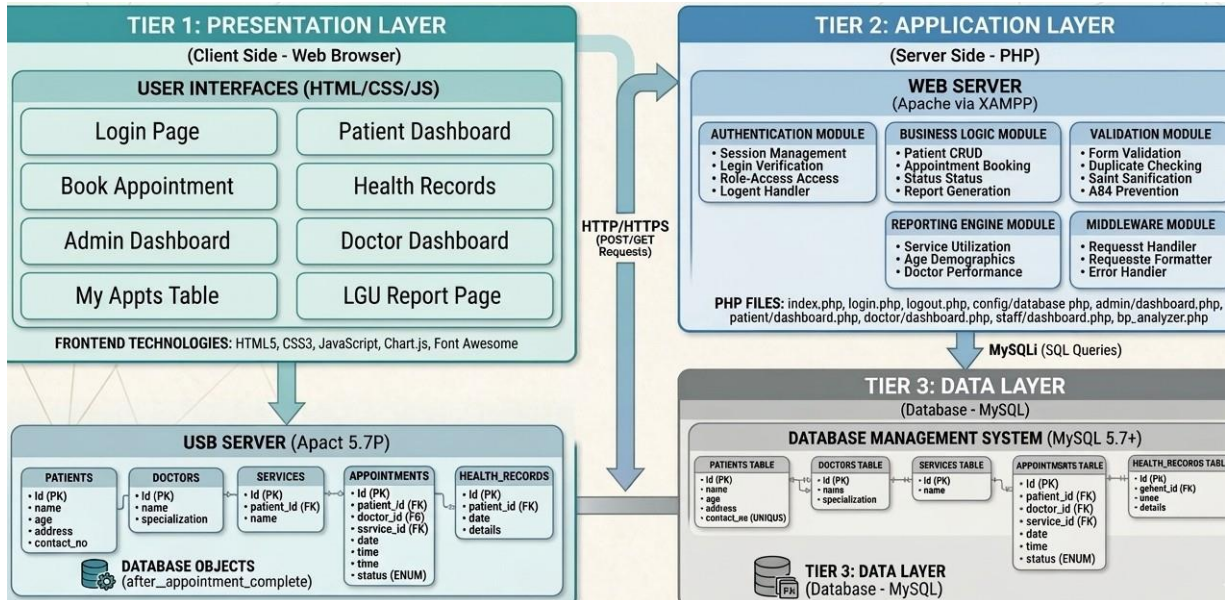
Section Description:

Data Architecture (ERD Diagram)

The Entity Relationship Diagram (ERD) defines the logical database structure of the Barangay Health Clinic system, enforcing data integrity and normalization. The PATIENTS entity tracks unique resident profiles to eliminate duplication, while the DOCTORS and SERVICES entities catalog healthcare providers and medical offerings, respectively. Linking these elements is the APPOINTMENTS entity, which manages transactional workflows like scheduling and status tracking under strict validation constraints to prevent double-bookings. Once a consultation is completed, the system automatically pipes data into the HEALTH_RECORD entity, maintaining an organized, historical timeline of diagnoses, symptoms, and medical notes per patient for clinical continuity.



System Infrastructure



Section Description: System Infrastructure (3-Tier Architecture)

The system infrastructure follows a localized 3-Tier Software Architecture engineered to isolate user interfaces, processing logic, and data persistence. The **Presentation Layer (Tier 1)** runs client-side in the web browser using HTML5, CSS3, JavaScript, to render tailored user dashboards and dynamic LGU analytical reports. Handling these client requests is the **Application Layer (Tier 2)**, powered by localized PHP scripts on an Apache server via XAMPP, which executes backend validation, role-based authentication, and core clinical business logic. Finally, the **Data Layer (Tier 3)** utilizes a local MySQL database to manage physical schemas and execute secure MySQL queries, ensuring strict transactional boundaries and protecting sensitive health records from unauthorized data tampering.



4. TELOS FEASIBILITY ANALYSIS

4.1 T - Technical Feasibility

Factor	Assessment	Justification
Development Tools	Feasible	XAMPP (free), VS Code (free), MySQL Workbench (free)
Programming Languages	Feasible	PHP, HTML, CSS, JavaScript - widely supported
Database	Feasible	MySQL 5.7+ (free, included in XAMPP)
Server Hardware	Feasible	Existing barangay computer (4GB RAM minimum)

4.2 E - Economic Feasibility

Cost Category	Amount	Details
Software Licenses	₱0	All software is open source/free
Development Hardware	₱0	Using existing barangay computer



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Server Hardware	₱0	Existing computer acts as server
Training Materials	₱0	Digital PDF manuals
Maintenance	₱0	Staff performs basic maintenance
Backup Storage	₱300 (one-time)	USB flash drive for backups
Total One-Time Cost	₱300	
Monthly Operational Cost	₱500	Estimated electricity

ROI Calculation:

- Staff time saved: 2-3 hours/day = ~₱700/month value
- Paperwork reduction: ₱200/month
- Payback Period: Less than 1 month

Verdict: Economically Feasible

4.3 L - Legal Feasibility

Legal Requirement	Status	Compliance Action



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Data Privacy Act (RA 10173)	Partial	Role-based access, session authentication implemented
Patient Consent	Required	Obtain written consent during registration (offline)
Medical Confidentiality	Compliant	Patient data isolated by patient_id filter
Open Source Licenses	Compliant	XAMPP, PHP, MySQL use open source licenses

Verdict: Legally Feasible (with proper consent forms)

4.4 O - Operational Feasibility

Factor	Assessment	Details
User Acceptance	High	Staff wants to replace manual logbooks



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Ease of Use	High	Simple web interface with clear buttons
Training Required	Low	1-2 hours of basic training
Workflow Integration	Positive	Replaces manual processes
Power Reliability	Stable	Barangay has stable electricity 8AM-5PM
Internet Dependency	Low	Works on local network

Verdict: Operationally Feasible

4.5 S - Scheduling Feasibility

Phase	Duration	Timeline
Requirements Gathering	7 days	Week 1



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Design (ERD, DFD, Architecture)	7 days	Week 2
Database Development (SQL)	3 days	Week 3
PHP Backend Development	7 days	Weeks 3-4
Frontend Development	5 days	Week 5
Testing & Bug Fixes	5 days	Week 6
Documentation	5 days	Week 7
Total	7 weeks	

Verdict: Scheduling Feasible

4.6 TELOS Summary



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Feasibility	Verdict
Technical	Feasible
Economic	Feasible
Legal	Feasible
Operational	Feasible
Scheduling	Feasible

UI/UX Blueprint





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Admin Dashboard:

BRGYClinic

Dashboard

Patients

Appointments

LGU Report

Doctors

Logout

Welcome, Administrator!

Administrator Dashboard

May 15, 2026

59

Total Registered Patients

15

Total Appointments

13

Completed

0

Pending

2

Cancelled

Recent Appointments

Search recent appointments...

Patient	Age	Doctor	Service	Date	Status	Action
Paolo Basas	15	Dr. Cruz	Pediatric	2026-05-13 10:00 AM	Completed	Change
Vhone	21	Dr. Santos	General Checkup	2026-04-29 01:00 PM	Completed	Change
Juan Dela Cruz	45	Dr. Santos	General Checkup	2026-04-27 04:00 PM	Completed	Change
Pedro Reyes	65	Dr. Santos	General Checkup	2026-04-27 09:00 AM	Completed	Change

Activate Windows
Go to Settings to activate Windows.

BRGYClinic

Dashboard

Patients

Appointments

LGU Report

Doctors

Logout

Welcome, Administrator!

Administrator Dashboard

May 15, 2026

Patient Management

Search patients by name, s

Add Patient

Add New Patient

Full Name

Age

Address

Contact Number

Add Patient Cancel

ID	Name	Age	Barangay	Contact Number	Action
72	Choloe Tagapan				
71	Paolo Basas				
65	Elizel Eco				
64	Rafael Sanchez				
58	MarthTumlos	13	Barangay 32	0912345	
57	Vhone	21	Tondo	0912345678	
53	Victor Rivera	72	Barangay 5	09123456750	
52	Urendo Castro	27	Barangay 4	09123456740	

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BRGYClinic

Dashboard

Patients

Appointments

LGU Report

Doctors

Logout

Welcome, Administrator!

Administrator Dashboard

May 15, 2026

All Appointments

All Status Search by patient or doctor

Patient	Age	Doctor	Service	Date	Time	Status	Action
Paolo Basas	15	Dr. Cruz	Pediatric	2026-05-13	10:00 AM	Completed	Change
Vhone	21	Dr. Santos	General Checkup	2026-04-29	01:00 PM	Completed	Change
Juan Dela Cruz	45	Dr. Santos	General Checkup	2026-04-27	04:00 PM	Completed	Change
Pedro Reyes	65	Dr. Santos	General Checkup	2026-04-27	09:00 AM	Completed	Change
Adriana Reyes	6	Dr. Santos	General Checkup	2026-01-06	09:30 AM	Completed	Change
Adriana Reyes	6	Dr. Santos	General Checkup	2025-12-30	09:30 AM	Completed	Change
Adriana Reyes	6	Dr. Santos	General Checkup	2025-12-25	09:30 AM	Completed	Change
Juan Dela Cruz	45	Dr. Santos	General Checkup	2025-12-23	10:00 AM	Completed	Change
Miguel Torres	40	Dr. Santos	General Checkup	2025-12-22	11:00 AM	Cancelled	Change

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BRGYClinic

Dashboard

Patients

Appointments

LGU Report

Doctors

Logout

Barangay Health Center - LGU Report

Year: 2026 | May 15, 2026 10:55 PM

Official Report for Local Government Unit (LGU)

15
Total Appointments

59
Registered Patients

13
Completed Consultations

86.7%
Success Rate

Service Utilization Report

Total Appointments

Completed

Cancelled

General Checkup

Total Appointments: 13

Completed: 12

Cancelled: 1

Number of Appointments

14

12

10

8

6

4

2

0

13

12

1

2

1

1

Activate Windows
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BRGYClinic

Dashboard

Patients

Appointments

LGU Report

Doctors

Logout

0

General Checkup

Pediatric Services

Prenatal

Service	Total	Completed	Cancelled	Pending	Confirmed	Patient Ages	Age Breakdown
General Checkup	13	12	1	0	0	6, 21, 32, 40, 45, 60, 65	Child: 3 Youth: 2 Adult: 5 Senior: 3
Pediatric	2	1	1	0	0	15, 32	Child: 0 Youth: 2 Adult: 0 Senior: 0
Prenatal	0	0	0	0	0	No data	Child: 0 Youth: 0 Adult: 0 Senior: 0
Total	15	13	2	0	0		

Doctor Service Summary

Doctor	Specialization	Service	Total Appointments	Completed	Success Rate
Dr. Santos	General Practitioner	General Checkup	13	12	92.3%
Dr. Reyes	Obstetrician	Prenatal	0	0	0%
Dr. Cruz	Pediatrician	Pediatric	2	1	50%

Export to CSV

Print Report

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BRGYClinic

Dashboard

Patients

Appointments

LGU Report

Doctors

Logout

Barangay Health Center - LGU Report

Year: 2023 | From: 01-01-2023 | To: 12-31-2023

LGU Report for Barangay Health Center - LGU Report

15

Total Appointments

59

Registered Patients

13

Completed Consultations

86.7%

Success Rate

Service Utilization Report

Bar chart showing Service Utilization Report with categories: General Checkup, Pediatric, Prenatal. Legend: Total Appointments (Blue), Registered Patients (Green), Completed Consultations (Red).

Barangay Health Center - LGU Report

Year: 2023 | From: 01-01-2023 | To: 12-31-2023

LGU Report for Barangay Health Center - LGU Report

Service	Total	Completed	Cancelled	Pending	Confirmed	Patient Ages	Age Breakdown
General Checkup	13	12	1	0	0	6, 21, 32, 40, 45, 60, 65	Child: 3 Youth: 2 Adult: 5 Senior: 3
Pediatric	2	1	1	0	0	15, 32	Child: 0 Youth: 2 Adult: 0 Senior: 0
Prenatal	0	0	0	0	0	No data	Child: 0 Youth: 0 Adult: 0 Senior: 0
Total	15	13	2	0	0		

Doctor Service Summary

Doctor	Specialization	Service	Total Appointments	Completed	Success Rate
Dr. Santos	General Practitioner	General Checkup	13	12	92.3%
Dr. Reyes	Obstetrician	Prenatal	0	0	0%
Dr. Cruz	Pediatrician	Pediatric	2	1	50%

Export to CSV

Print Report

Activate Windows
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Print

2 pages

Destination: Save as PDF

Pages: All

Layout: Landscape

More settings

Save Cancel

Activate Windows
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NATIONAL TEACHERS COLLEGE

BRGYClinic

Dashboard

Patients

Appointments

LGU Report

Doctors

Logout

Welcome, Administrator!

Administrator Dashboard

May 15, 2026

Doctors & Services

Doctor Name	Specialization	Service	Schedule
Dr. Santos	General Practitioner	General Checkup	Mon-Fri, 9AM-5PM
Dr. Reyes	Obstetrician	Prenatal	Tue-Thu, 10AM-4PM
Dr. Cruz	Pediatrician	Pediatric	Mon-Wed, 8AM-3PM

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ADMIN DASHBOARD

The Admin Dashboard is the most feature-rich interface, giving the administrator full control over the system. The left sidebar contains navigation links to Dashboard, Patients, Appointments, LGU Reports, and Doctors. The main area displays five statistic cards showing total patients, total appointments, completed, pending, and cancelled counts. A bar chart visualizes patient demographics by age group. The Patients section allows full CRUD operations with add, edit, delete, and search functionality, including duplicate validation to prevent duplicate records. The Appointments section shows all appointments with status badges and allows the admin to update status from Pending to Confirmed, Completed, or Cancelled. The LGU Reports section generates official health reports including service utilization tables with age breakdowns, doctor performance summaries, and CSV export or print options. Every action provides green success or red error feedback messages.

Patient Dashboard:

Paolo Basas
Age: 15

Dashboard

My Appointments

Book Appointment

Health Records

Logout

Welcome, Paolo Basas!

May 15, 2026

1
Total Appointments

0
Pending

0
Confirmed

1
Health Records

Recent Appointments

Date	Time	Doctor	Service	Status	Remarks
2026-05-13	10:00 AM	Dr. Cruz	Pediatric	Completed	ang sakit po ng ulo tuwing gabi [Doctor Note: pwed

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Paolo Basas
Age: 15

Dashboard

My Appointments

Book Appointment

Health Records

Welcome, Paolo Basas!

May 15, 2026

All My Appointments

Date	Time	Doctor	Service	Status	Reason/Remarks
2026-05-13	10:00 AM	Dr. Cruz	Pediatric	Completed	ang sakit po ng ulo tuwing gabi [Doctor Note: pwede kana po pumunta]



Paolo Basas
Age: 15

Dashboard

My Appointments

Book Appointment

Health Records

Welcome, Paolo Basas!

May 15, 2026

Book New Appointment

Select Service

General Checkup (Dr. Santos)

Preferred Date

05 / 21 / 2026

Preferred Time

3:00 PM

Reason for Appointment

Iumalabo po ang mata

Submit Appointment Request

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Paolo Basas
Age: 15

Dashboard

My Appointments

Book Appointment

Health Records

Welcome, Paolo Basas!

May 15, 2026

My Health Records

Date	Details
2026-05-13	Appointment completed: ang sakit po ng ulo tuwing gabi [Doctor Note: pwede kana po pumunta]

PATIENT DASHBOARD

The Patient Dashboard is the most user-friendly interface, allowing registered residents to manage their healthcare journey. The sidebar provides access to Dashboard, My Appointments, Book Appointment, and Health Records. The Book Appointment section contains a simple form where patients select a service from three options General Checkup, Prenatal, or Pediatric, pick a future date using a date picker, select a time from hourly slots, and enter their reason for visit. When submitted, the system validates doctor availability, day of week, working hours, and double booking, then saves the appointment with "Pending" status. The My Appointments section displays all appointments with color-coded status badges: yellow for Pending, green for Confirmed, blue for Completed, and red for Cancelled. The Health Records section shows all medical records automatically generated from completed appointments



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Doctor Dashboard:

Dr. Santos
Doctor

Dashboard

Appointments

My Schedule

Logout

Welcome, Dr. Santos!

May 15, 2026

14
Total Appointments

1
Pending

0
Confirmed

12
Completed

Today's Schedule

No appointments scheduled for today.

Activate Windows
Go to Settings to activate Windows.

Dr. Santos
Doctor

Dashboard

Appointments

My Schedule

Logout

Welcome, Dr. Santos!

May 15, 2026

All Appointments

Patient	Age	Service	Date	Time	Status	Notes	Action
Paolo Basas	15	General Checkup	2026-05-20	03:00 PM	Pending	Iumalabo po ang mata ko	No Action
Vhone	21	General Checkup	2026-04-29	01:00 PM	Completed	ang sakit ng likod ko kaka code [Doctor Note: bumili ka kase]	No Action
Juan Dela Cruz	45	General Checkup	2026-04-27	04:00 PM	Completed	back hurt [Doctor Note: try to rest and sleep well] [Doctor]	No Action
Pedro Reyes	65	General Checkup	2026-04-27	09:00 AM	Completed	my legs hurt and my back [Doctor Note: matulog kapo]	No Action
Adriana Reyes	6	General Checkup	2026-01-06	09:30 AM	Completed	checkup	No Action
Adriana Reyes	6	General Checkup	2025-12-30	09:30 AM	Completed	checkup	No Action
Adriana Reyes	6	General Checkup	2025-12-25	09:30 AM	Completed	checkup	No Action
Juan Dela Cruz	45	General Checkup	2025-12-23	10:00 AM	Completed	body checkup	No Action
Miguel Torres	40	General Checkup	2025-12-22	11:00 AM	Cancelled	Pediatric checkup	No Action
Maria Santos	32	General Checkup	2025-12-19	10:00 AM	Completed	Test appointment	No Action

Dr. Santos
Doctor

Dashboard

Appointments

My Schedule

Logout

Welcome, Dr. Santos!

May 15, 2026

My Working Schedule

Working Days: Monday - Friday

Working Hours: 9:00 AM - 4:00 PM



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DOCTOR DASHBOARD

The Doctor Dashboard is designed for medical professionals to manage their patient consultations. Upon login, the doctor sees four statistic cards showing total appointments, pending, confirmed, and completed counts. The Today's Schedule section lists all patients scheduled for the current date with their appointment time, name, age, service, and status. For confirmed appointments, a blue "Add Note" button opens a modal form where doctors can enter medical notes, diagnosis, and prescriptions. The Appointments section displays the complete historical list of all appointments assigned to the doctor, with notes visible in a dedicated column. The My Schedule section shows the doctor's official working days and hours in a highlighted box. For Dr. Santos it is Monday-Friday 9AM-5PM, for Dr. Reyes it is Tuesday-Thursday 10AM-4PM, and for Dr. Cruz it is Monday-Wednesday 8AM-3PM.

Staff Dashboard:

Brgy Clinic

Dashboard

LGU Report

Logout

Welcome, Barangay Staff!

Barangay Health Staff - Resident Management & LGU Reports

May 15, 2026

59

Total Residents

18

Senior Citizens (60+)

11

Children (≤12)

13

Completed Consults

Complete Resident Directory

Search by name, address, or contact number...

ID	Full Name	Age	Address	Contact Number	Category
72	Choloe Tagapan	10	Cavite 3590	09111	Child
71	Paolo Basas	15	barangay 1031	09123	Adult
65	Eizel Eco	29	CM.Recto Tondo manila	0935	Adult
64	Rafael Sanchez	57	numancia brgy 143	09123789	Adult

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Brgy Clinic

Dashboard

LGU Report

Logout

Barangay Health Center - LGU Report

Year: 2026 | May 15, 2026 11:03 PM

Official Report for Local Government Unit (LGU)

Prepared by: Barangay Staff (Barangay Health Staff)

16

Total Appointments

59

Registered Patients

13

Completed Consultations

81.3%

Success Rate

Service Utilization Report

Total Appointments

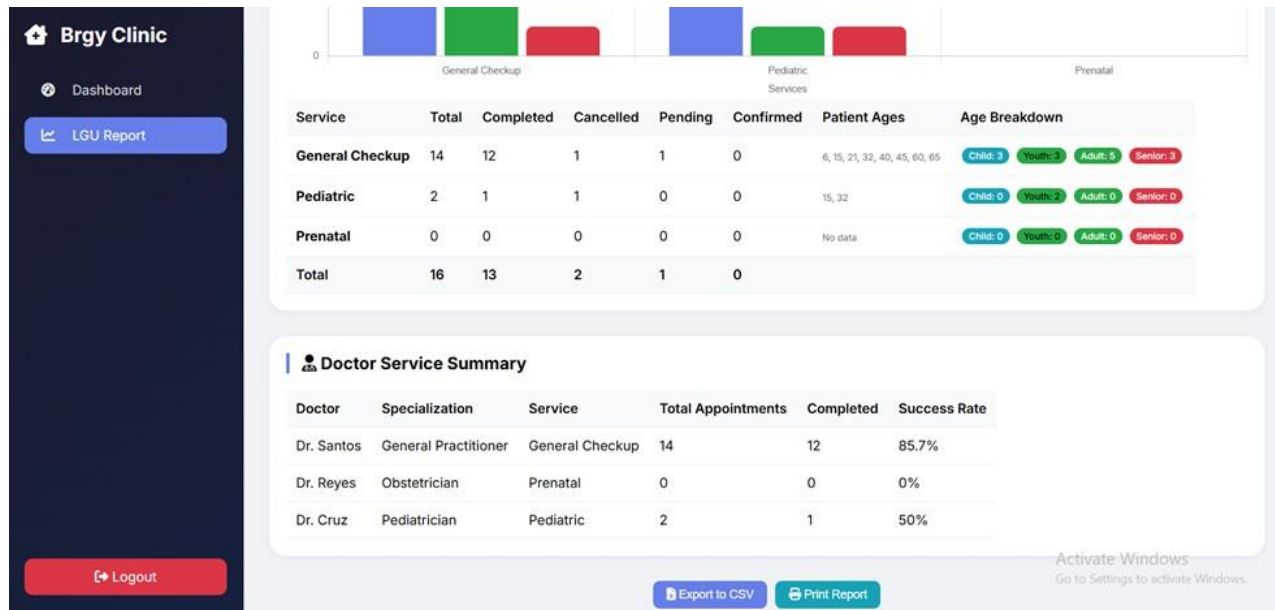
Completed

Cancelled

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STAFF DASHBOARD

The Barangay Staff Dashboard is simplified for health workers who need resident visibility and reporting capabilities without full admin rights. The dashboard displays four statistic cards showing total residents, senior citizens aged sixty and above, children aged twelve and below, and completed consultations. Below the statistics, the Complete Resident Directory shows all registered patients in a searchable table with columns for ID, name, age, address, contact number, and a color-coded category badge red for Senior, blue for Child, green for Adult. The LGU Report section provides the same reporting features as the admin version, including exclusive summary, service utilization table with age breakdowns, doctor performance summary, and CSV export or print options for submission to the Local Government Unit.



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6.1 Requirements to Test Cases Traceability Matrix

REQ ID	Requirement	Test Case ID	Test Case Description	Status
FR-01	Prevent double booking	TC-01	Attempt to book same doctor at same date/time → Should show error	Pass
FR-02	Validate doctor schedules	TC-02	Book appointment outside doctor's hours → Should reject	Pass



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FR-03	Eliminate duplicate patient records	TC-03	Add patient with existing contact number → Should show error	Pass
FR-04	Generate summary reports	TC-04	Click LGU Report → Should display service utilization table	Pass
FR-05	Validate appointment inputs	TC-05	Submit empty form → Should show validation errors	Pass
FR-06	Enforce doctor-service rules	TC-06	Book General Checkup with Dr. Reyes → Should reject	Pass
FR-07	View health records	TC-07	Patient clicks Health Records → Should display records	Pass
FR-08	Prevent duplicate patients	TC-09	Add exact same patient twice → Should show duplicate error	Pass



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FR-9	Service utilization reports	TC-10	View LGU Report Should show counts per service	Pass
FR-10	Update appointment status	TC-11	Admin changes status Should update and show success	Pass
FR-11	Auto-create health record	TC-12	Change status to Completed Health record auto-created	Pass
FR-12	Add medical notes	TC-13	Doctor adds note Note appended to appointment	Pass
FR-13	Search functionality	TC-14	Search by name Shows matching patients only	Pass
FR-14	Role-based access control	TC-15	Patient tries admin URL Redirects to login	Pass



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6.2 Detailed Test Cases

Test Case TC-01: Prevent Double Booking

Test Field	Value
Test ID	TC-01
Requirement	FR-01
Precondition	Doctor has existing appointment on 2026-05-20 at 10:00 AM
Test Steps	1. Login as Patient 2. Book appointment for same doctor on 2026-05-20 at 10:00 AM 3. Click Submit
Expected Result	Error message: "Doctor already booked at this time"
Actual Result	Error message displayed
Status	PASS



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Test Case TC-02: Validate Doctor Schedule

Test Field	Value
Test ID	TC-02
Requirement	FR-02
Precondition	Dr. Santos works Mon-Fri, 9AM-5PM
Test Steps	1. Login as Patient 2. Select General Checkup (Dr. Santos) 3. Select Sunday date at 10:00 AM 4. Click Submit
Expected Result	Error: "GP available Mon–Fri, 9AM–5PM only"
Actual Result	Error message displayed
Status	PASS